

Data Quality, Audit & Cleaning Report

Dynamic Delivery Time Prediction System — Brazilian E-Commerce

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1. Executive Summary

This report documents the rigorous data understanding, audit, and cleaning procedures applied to the Olist Brazilian E-Commerce dataset. The objective was to transform raw relational transactional tables into an order-level, leak-free, model-ready dataset to predict delivery duration (in days).

2. Raw Relational Tables Audit

Table Name	Raw Records	Primary Key	Key Attributes
orders	99,441	order_id	status, purchase_date, delivered_date
order_items	112,650	order_id + item_id	product_id, seller_id, price, freight
products	32,951	product_id	category_name, weight_g, volume
sellers	3,095	seller_id	seller_zip, seller_city, seller_state
customers	99,441	customer_id	customer_zip, customer_city, customer_state

3. Relational Integrity & Cleaning Rules

Automated checks validated 100% uniqueness of primary keys and confirmed 0 orphan foreign key records in order_items. Non-delivered orders, missing delivery dates, and negative durations were removed. Missing product dimensions were dropped, and missing category names imputed as 'unknown'.

4. Leak-Free Feature Engineering & Final Aggregation

Features were restricted to checkout-available data: intra-state delivery indicators, leave-one-out historical seller delivery averages (preventing target leakage), package volumes, and basket sizes. 110,178 item records were aggregated to **96,460 clean order rows** with zero missing values.